Unit 5 – Polynomial Functions – Day 5 *Accelerated II/III*

Dividing Polynomials – Synthetic Division

**ONLY works with LINEAR divisors!!

Example 1: Divide $x^3 - 3x^2 - 7x + 6$ by $x - 4$

Example 2: $\left(2x^3 - 3x^2 + 4x + 5\right) \div \left(x + 2\right)$

Example 3: $\left(x^3 - 2x^2 - 4\right) \div \left(x - 3\right)$

Example 4: $\left(3x^3 + 14x^2 - 28x - 24\right) \div \left(x + 6\right)$

Example 5: Use synthetic division to find

A) $f(-4)$ for $f(x) = 3x^3 + 13x^2 + 2x - 8$

B) $f(3)$ for $f(x) = 4x^3 - 6x - 4$